Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	17 October 2026
Team ID	SWUID2026403471
Project Name	IMovies
Maximum Marks	4 Marks

1. Functional Requirements

Functional requirements describe what the system must do.

ID	Module	Requirement	Priority
FR-01	User Authentication	The system shall allow users to register, log in and log out securely, and to reset or change their password.	High
FR-02	Movie Browsing	The system shall display Now Showing and Upcoming movies with details (synopsis, cast, trailer, ratings) and allow search and filtering by title, language, genre and rating.	High
FR-03	Show Selection	The system shall allow users to select a city and view theatres, dates and showtimes for a movie.	High
FR-04	Seat Selection	The system shall display an interactive seat map, allow selection of multiple seats and temporarily hold them during payment.	High
FR-05	Booking & Payment	The system shall show a booking summary, allow coupon codes, accept payment through UPI, cards and net banking, and confirm the booking only after successful payment.	High
FR-06	E-Ticket & Notification	The system shall generate an e-ticket with a booking ID and QR code and send it with a confirmation by email/SMS.	High
FR-07	Booking History	The system shall allow users to view their past and upcoming bookings and e-tickets.	Medium
FR-08	Cancellation & Refund	The system shall allow users to cancel a booking as per the cancellation policy and refund the amount to the original payment method.	High
FR-09	Reviews & Ratings	The system shall allow users who booked a movie to rate and review it, and display ratings and reviews on the movie page.	Medium
FR-10	Admin Management	The system shall allow the admin to manage movies, theatres, shows, ticket prices and coupons, and view booking and sales reports.	High

2. Non-Functional Requirements

Non-functional requirements describe how well the system must perform and the constraints it must meet.

ID	Category	Requirement	Measure / Criteria
NFR-01	Performance	Pages shall load within 3 seconds, search results within 2 seconds, and seat availability shall update in near real time.	Response time
NFR-02	Scalability	The system shall support at least 1,000 concurrent users, including peak load on new movie releases.	Concurrent users
NFR-03	Availability	The system shall be available 99.5% of the time, excluding planned maintenance.	Monthly uptime
NFR-04	Reliability	A seat shall never be booked twice, and a failed payment shall not confirm a booking.	Zero double bookings
NFR-05	Security – Data	Passwords shall be stored hashed, all data shall travel over HTTPS, and sessions shall expire after inactivity.	Hashing, HTTPS, timeout
NFR-06	Security – Payment	Card details shall not be stored by the system; payments shall go through a PCI-DSS compliant gateway.	Payment compliance
NFR-07	Security – Access	The system shall protect against SQL injection, XSS and CSRF, and use role-based access for users and admin.	Input validation, RBAC
NFR-08	Usability	The interface shall be simple, accessible (readable fonts, good contrast) and let a new user complete a booking in 5 steps or fewer.	Ease of use
NFR-09	Compatibility	The website shall be responsive and work on current versions of Chrome, Firefox, Edge and Safari.	Cross-browser, responsive
NFR-10	Maintainability & Data Integrity	The code shall be modular and documented, and booking and payment data shall be consistent with daily database backups.	Modular design, daily backup